Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

SYSTEM ON CHIP DESIGN [ECE 4087]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) A) Company XYZ, a semiconductor IP provider, is tasked with developing an advanced image processing IP for a new flagship smartphone company. The smartphone manufacturer requires higher AI-based image processing performance with low power consumption, a small die area, and quick time-to-market compatibility. What are the key trade-offs between performance, power, and area in designing this IP, and how can Company XYZ address them? Should Company XYZ license this IP exclusively or make it broadly available? What kind of IP XYZ company should aim for - Hard, Firm or Soft. Outline the pros and cons. (5)

B) Formulate an assembly language (Little Man or MARIE) program to find the smallest of the 3 numbers input by the user and sent it to the output. Draw a flowchart to help yourself with the proper implementation of the problem. (3)

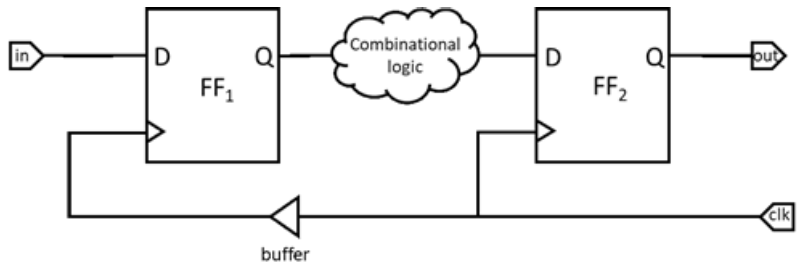
C) A five-stage pipelined processor has the stage delays as 250, 220, 160, 110 and 190 ps respectively. Registers are used between the stages and have a delay of 12 ps each. Assuming constant clocking rate, Compute the total time taken to process 10000 data items on the pipeline. Demonstrate the improvement in speed-up if any two stages can be split into smaller two stages with equal delays. (2)

2) A) As a System-on-Chip architect design an efficient datapath for a High-Speed Intellectual Property (IP) module to output the approximate the square root of any number input to the module, utilizing registers, adders/subtractors, comparators, and other standard/basic digital blocks only. (5)

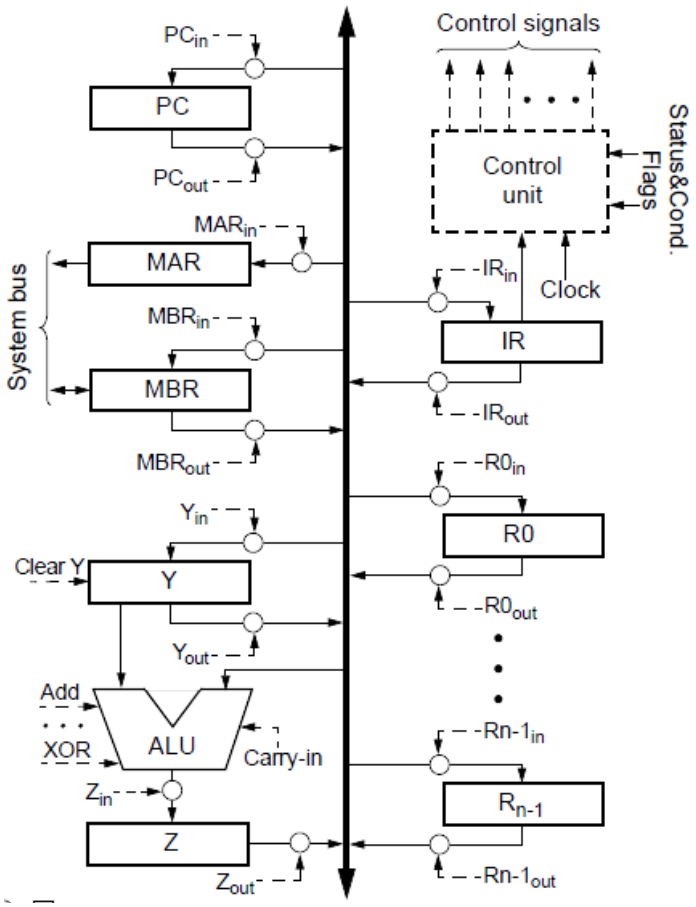
B) Construct a state diagram to represent the controller of the Square Root (IP) module to output the approximate the square root of any number input it. Illustrate the different states and control outputs for each state in the FSM. (3)

C) Explain difference between Amdahl's law and Gustafson Law? Write equations and illustrate both laws with suitable examples. (2)

3) A) Find the maximum possible clock frequency for the given circuit in the Figure below to operate without failure. The delays and set-up times are given below. Delay values of Flipflop and Combinational circuit are $t_{_setup}$= 5ns and $t_{_hold}$= 4ns. The delay of the buffer is $t_{_buf}$= 2ns, the delay of combinational logic and flipflop are **t**_clk-q_max=5ns, **t**_clk-q_min=4ns, **t**_comb_max=10ns, **t**_comb_min=5ns. Write the hold time constraint equation. What is the maximum allowable jitter for the circuit to work in this frequency if $t_{_buf}$ is neglected? (4)



B) Conceptualize the control path of a new Memory compute instruction '**ADDMEM Rx, Ry, Rz'** (ADD two data stored in Memory and store result back in Memory) ADD data stored in Memory locations pointed by **Ry** and **Rz** and store the result in Memory address pointed by **Rx, e.g.** ADDMEM R1, R2, R3). Explain in terms of RTL (Micro-operations) and changes in various Registers like MAR, MBR, AC, PC, IR during fetch-decode-execute cycles. (use Figure for reference).

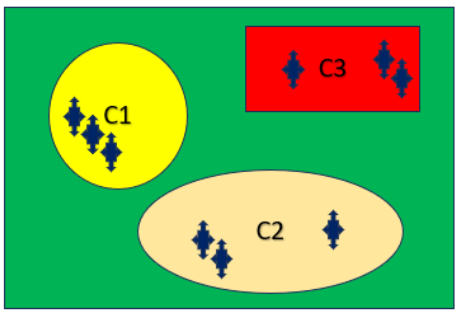


(3)

C) Illustrate how can the decision tree algorithm be formulated to recommend the clock domains for a new sequential module based on features such as path (Data or Control), power, and delay? The available data is given in the Figure. The new data which needs recommendation are given below

D1 (C, 20,2.5), D2 (D, 40,8.5), D3 (C, 18,7), D4 (D, 29, 1).

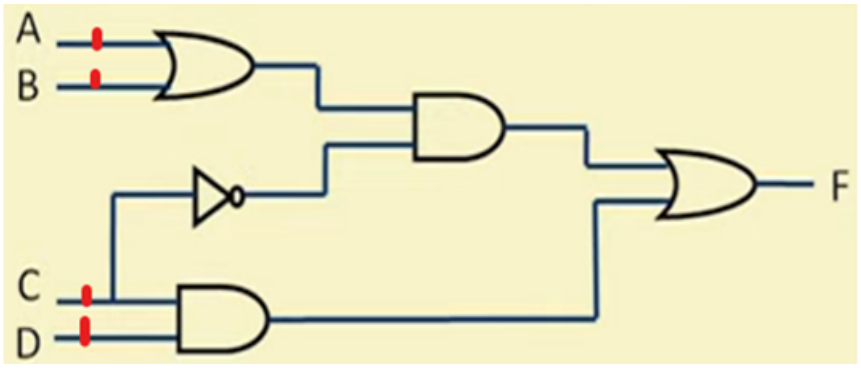
Datapath (D) or Controlpath (C)	Power Dissipation (nJ)	Delay (ps)	Clock Domain (1,2,3)
D	15	6.6	**1**
D	25	4.0	**2**
C	32	3.1	**3**
D	35	3.25	**2**
C	12	8.33	**1**
C	14	7.2	**1**
D	30	3.33	**2**



(3)

4) Find the minimum number of Test vectors to detect stuck at faults (both 0 and 1) at the input of the circuit shown in Figure at the following input points (i) A (ii) B (iii) C and (iv) D. (4)

A)

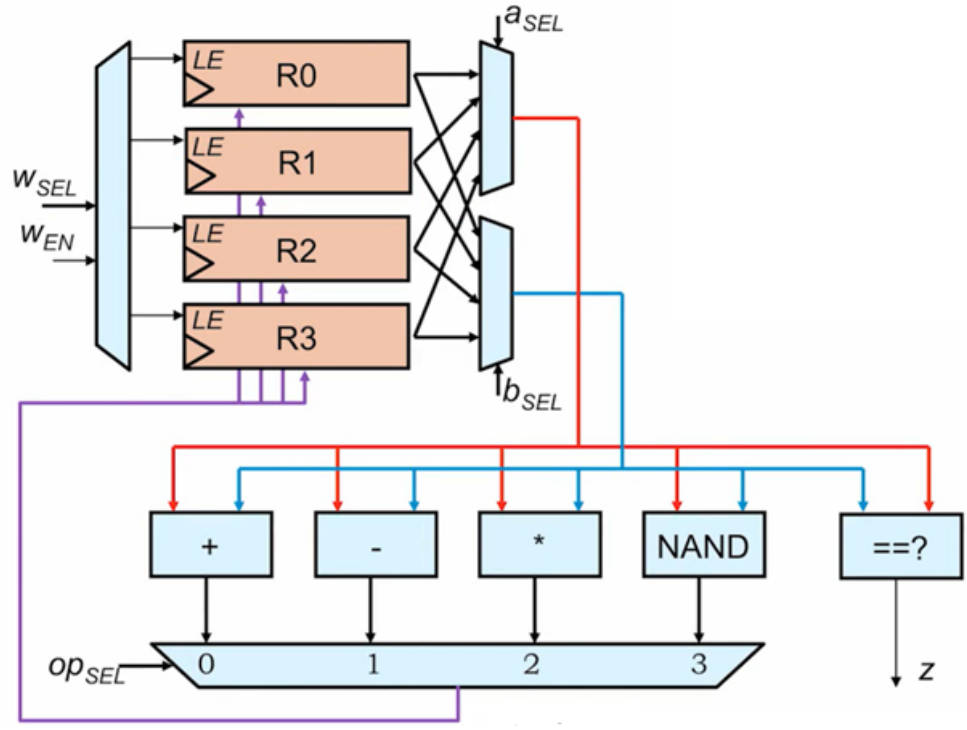


B) You are tasked with verifying a complex processor core using Universal Verification Methodology (UVM). With the help of a block diagram illustrate the key components of a UVM-based testbench. What is the role of each component and how do they communicate and coordinate the generation and collection of stimulus and responses for the Design Under Test (DUT)? (4)

C) In SoC cell placement optimization, both Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) are popular techniques for minimizing wire length, area and timing. Demonstrate and compare the particle/ position update mechanism in these two algorithms. How do these mechanisms help each algorithm explore and refine potential solutions? (2)

5) Design the control flow and control signals required to configure this programmable Datapath given in the Figure. 5A to perform a modulo operation (Rem=a%b. To find remainder when a given dividend is divided by a divisor). You may add more registers if required. Assume missing data and state them?

A)



(4)

B) At the IP scheduling level, analyze and determine the average waiting time, turnaround time, and response time for a Dual-core processor using the following scheduling strategies: 1. First-Come-First-Serve, 2. Shortest Job First, and 3. Round Robin with a time quantum of 2 units. Refer to Table below for below for the details of the task. (4)

Table 1

Process	Arrival time	Burst Time
P1	6	6
P2	4	8
P3	2	4
P4	8	8
P5	8	4
P6	10	2
P7	6	5
P8	2	3
P9	5	7
P10	1	5

C) As a Multicore Architect, you are tasked with redesigning the controller for an octa-core system to count the number of prime numbers between two large numbers, N1 and N2. Outline your considerations to ensure optimal core utilization and address potential cache coherency challenges. (2)

-----End-----